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Rajayakshma and Shosha

- ⇒ To learn the concept of Rajayakshma
- ⇒ To learn the Nidana of Rajayakshma mentioned in Ayurveda literatures
- ⇒ To frame out the probable Samprapti Ghataka based on the progression of the disease
- ⇒ To describe Vyadhi Svabhava (cardinal feature) of Rajayakshma
- ⇒ To know about the types of Rajayakshma
- ⇒ To learn about the features of Tri, Shad and Ekadasharupi Rajayakshma
- ⇒ To learn about Sadhyasadhyata of Rajayakshma
- ⇒ To classify and describe Shosha

*Rajayakshma*¹ is the spectrum of diseases where one can observe extreme emaciation of the body of the patient, depletion of all body tissues. It usually involves *Pranavaha Srotas*, This is mainly caused due to the vitiation of all three *Dosha*. As per *Charaka*, *Rajayakshma* is considered to be one among the *Rogasamuha* (a group of diseases). This is also called as *Shosha* (disease where dryness of all tissues take place), *Kshaya* (disease where depletion of body tissues), *Rogarat* (king of all diseases).

राजयक्ष्मा क्षय शोषो रोगराडिति च स्मृतः॥१॥

Rajayakshma is considered to be associated with multiple diseases which occur prior to the manifestation of *Rajayakshma* and multiple diseases will follow *Rajayakshma*. That is the reason, *Rajayakshma* is considered to be the king of the diseases or one of the major diseases.

नक्षत्राणा द्विजाना च राशोऽभूद्यदयं पुरा।

यच्च राजा च यक्ष्मा च राजयक्ष्मा ततो मत॥२॥

देहौषधक्षयकृतेः क्षयस्तत्सम्भवाच्च स।

- 1 The main pathogenesis in this disease is depletion of all body tissues from the body. In other words, the patient needs much more time to get recovery from the illness, as the treatment of the patient is taking much longer time as there is sufficient nutrient loss is already present in the patient and duration of the illness is prolonged.
- Hampered functions of all types of Agni – Jatharagni, Dhatvagni and Bhutagni
 - All the three Doosha affliction can be observed.
 - All the seven Dhatu affliction can be seen.
 - All the three Rogamarga affliction can be seen.
 - Lakshana observed in the patient implies affliction of the multiple organs in the body.
 - The treatment as per Ayurveda fundamentals, is facilitating the movement of Dosha from the Shakha or Madhyama Rogamarga to the Koshtha. This is difficult as the Rogibala in this patient will be already weaker and extreme Dhatu Kshaya is seen in the patient.
- As per Ayurveda, human body has nine natural orifices in males and in females, twelve natural orifices are seen. Due to the exposure of the aetiological factor, each of these orifices are susceptible to the entry of the pathogen into the body leading to multiple disease manifestations. Rajayakshma is said to be manifested in the individuals due to Sahasa (excess exercise), Vishamashana (abnormal diet pattern), Vega Sandharana (suppression of the natural urges) and Kshaya (excess sexual intercourse). In case of extreme Sahasa (exercise), the first and foremost orifice affliction can be observed is nose and oral cavity i.e. respiratory system and alimentary canal. In case of Vishamashana, primarily affliction of alimentary canal will be seen. In case of Kshaya, Shukra Kshaya in particular, primary affliction of lower alimentary canal can be seen. In case of Kshaya, Shukra Kshaya in particular, primary affliction of reproductive and urinary system can be observed. Thus, we have to analyse each type of Nidana exposure in the manifestation of Rajayakshma.

रसादिशोषणाच्छोषो रोगराट् तेषु राजनात्॥३॥

A.Hr.Ni. - 5/2-3

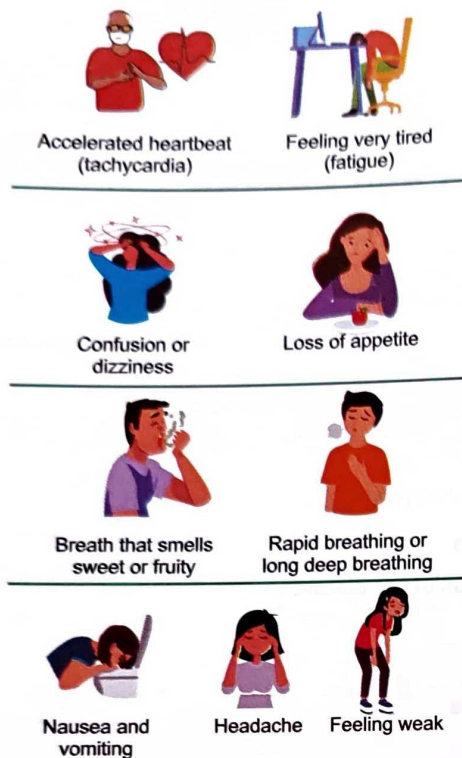


Figure 29: Lakshana of Rajayakshma

- ① The disease afflicted the king of stars, Chandra (moon) and it is considered as king of diseases and it is having large number of the symptoms, it is called *Rajayakshma*.
- ② The disease leads to loss or depletion of body tissues, this is called as *Kshaya*.
- ③ As all *Dhatu* undergo depletion from *Rasa* to *Shukra Dhatu*, this is also called as *Kshaya*.

Nidana² of Rajayakshma

इह खलु चत्वारि शोषस्यायतनानि भवन्ति; त-

द्यथा - साहसं संधारणं क्षयो विषमाशनमिति॥३॥

Ca.Ni. - 6/3

अयथाबलमारम्भं वेगसंधारणं क्षयम्।

यक्ष्मणः कारणं विद्याच्चतुर्थं विषमाशनम्॥

Ca.Ci. - 8/13

As far as *Rajayakshma* is concerned, the disease is classified into 4 based on the *Nidana* or *Ayatana* or causative factors are concerned.

1. *Sahasajanya*³ (over exertion)
2. *Vegasandharanajanya*⁴ (suppression of the urges)
3. *Kshayajanya*⁵ (depletion of the body tissues)
4. *Vishamashanajanya*⁶ (improper diet induced)

- 2 When all the *Nidana* mentioned in *Rajayakshma* are analysed, we are going to observe the sequential depletion of carbohydrates, fats, proteins, water and micronutrients such as minerals, vitamins etc. Depending upon the degree of the depletion of body tissues, the patient will develop the clinical manifestation of symptoms. If the person is having such a depletion of body tissues, then he is not in a position to produce the cells including RBC, WBC, platelets etc. The person is more susceptible for any kind of opportunistic infections. These opportunistic infections can occur because of the entry of organisms through any of the natural orifices.
- 3 When a person indulges over exertion and he is not consuming adequate amount of nutrition, he will likely to have sequential decrease of macronutrients such as carbohydrates, fats, proteins and water; micronutrients such as minerals and vitamins. As there is reduced nutrient supply, new cells are not formed; including deficiency in formation of RBCs, WBCs and platelets. Thus, the person is exposed to multiple opportunistic infections.
- 4 The term *Vegasandharana* refers to suppression of natural urges which should not be suppressed. Let us take few examples as description of all these urges is a huge task and makes this book a gigantic one.
Let us take the example of *Purisha Vega Dharana* (suppression of urge of defaecation). When the person suppresses the urge of defaecation, that will lead to the more solidification of stool by absorption of 75% of water present in the stool usually. Further, this will lead to stasis of faecal matter for much longer time. This will become a source of bacterial infection to proliferate in the colon.
Let us take the examples of *Kasa* (cough), *Shvasa* (breathlessness due to exertion) due to and *Kshavathu* (sneezing). All these natural urges are leading to the compromise in the respiratory function. Cough is a protective mechanism which expels the allergens from the upper and lower respiratory passages to the exterior. Breathlessness due to exertion will be responsible for breathing through open mouth which may lead to the entry of the multiple pathogens into the respiratory airways leading to the disease manifestation. *Kshavathu* or sneezing is observed at the beginning of entry of allergens into the upper respiratory passages. If this urge is suppressed, allergens or pathogens will enter into the lower respiratory airways leading to the manifestation of disease in the lower respiratory passages.
- 5 In this cause, the person may already have the depletion of the body tissues; it implies he will have deficiency of macronutrients such as water, carbohydrates, fats and proteins and micronutrients such as minerals and vitamins. Depending upon the nature of depletion of body tissues, the person may show clinical symptoms.
- 6 The term *Vishamashana* refers to abnormal intake of food in terms of quantity and timing. In terms of quantity, sometimes the person will take too much amount of food and in some occasions, he will take too little amount of food. In terms of timing, the person will take the food too early and sometimes he will take the food too late after several hours getting the appetite. Here, the condition is the person is not following a particular ritual or he is not disciplined in terms of consumption of food. If the person is indulging in *Vishamashana* for longer duration, then only the person may get afflicted with *Rajayakshma*.
If the person is consuming the food too early, it shows the pathological effects similar to *Adhyashana* (consumption of food prior to digestion of previously ingested meal). Refer the context of *Agnimandya*. If the person is consuming the food too late, the nutrient supply becomes delayed because the person is having nutrient loss happened already in the body. In other words to say, the person is having glucose depletion in the beginning. As the person consumes the food after several hours of appearance of appetite, that will lead to the depletion of fats, proteins, water and deficiency of minerals and vitamins. If the person is consuming too little amount of food, then the same mechanism of consumption of food too late can be observed in the patients.

Samprapti of Rajayakshma

रसः स्रोतःसु रुद्धेषु स्वस्थानस्थो विद्यते।

स ऊर्ध्वं कासवेगेन बहुरूपः प्रवर्तते॥४३॥

कफप्रधानदपैस्तु रुद्धेषु रसवर्त्मसु।

अतिव्यबाधिनो वाऽपि क्षीणे रेतस्यनन्तराः।

क्षीयन्ते धातवः सर्वे ततः शुष्यति मानवः॥४२॥

Ca.Ci. – 8/43

Ma.Ni. – 10/2

• *Trido* – **Samprapti Ghtaka**

• **Dosha** : *Kapha* dominant *Tridosha*

• **Dushya** : All seven *Dhatu*

• **Srotas** : *Pranavaha* and *Rasavaha Srotas*

• **Srotodushti Prakara** : *Sanga* and *Vimargagamana*

• **Agni** : *Jatharagnimandya* and *Saptadhatvagnimandya*

• **Ama** : *Jatharagnimandya* and *Saptadhatvagnimandya*

• **Udbhava Sthana** : *Amashaya*

• **Sanchara Sthana** : *Sarva Sharira*

• **Vyakta Sthana** : *Uras*

• **Rogamarga** : *Abhyantara, Bahya* and *Madhyama*

• **Vyadhyavastha** : *Ashukari*

• **Sadhyasadhya** : *Asadhyata*

Purvarupa of Rajayakshma

तस्येमानि पूर्वरूपाणि भवन्ति; तद्यथा- प्रतिश्यायः,

क्षवथुरभीक्षणं, श्लेष्मप्रसेकः, मुखमाधुर्यम्, अनन्नाभिलाषः, अन्नकाले चायासः, दोषदर्शनमदोषेष्वल्पदोषेषु वा भावेषु पात्रोदकाभ्रसूपापूपोपदंशपरिवेशकेषु, भुक्तवतश्चास्य हल्लासः, तथोल्लेखनमप्याहारस्यान्तरान्तरा, मुखस्य पादयोश्च शोफः पाणयोश्चावेक्षणमत्यर्थम्, अक्ष्णोः श्वेतावभासता चातिमात्रं, बाह्योश्च प्रमाणजिज्ञासा, स्त्रीकामता, निर्धृणित्वं, बीभत्सदर्शनता चास्य काये, स्वप्ने चाभीक्षणं दर्शनमनुदकानामुदकस्थानानां शून्यानां च ग्रामनगरनिगमजनपदानां शुष्कदग्धभग्नानां च वनानां कृकलासमयूरवानरशुकसर्पकाकोलूकादिभिः संस्पर्श नमधिरोगणं यानं वा श्रोष्ट्रखरवराहैः केशास्थिभस्मतुषाङ्गाराशीनां चाधिरोहणमिति (शोषपूर्वरूपाणि भवन्ति)॥१३॥ Ca.Ni. – 6/13

• Running nose⁷

• Frequent sneezing

• Excess mucus mixed salivation⁸

• Sweetish taste in the mouth

• Lack of desire to take food

• Exhaustion during the time of consumption of food

• Finding fault in any of the food or food preparations even though the fault is mild

• Nausea

• Frequent vomiting⁹

• Swelling in the oral cavity and both feet¹⁰

• Frequently seeing one's hands¹¹

7 In this situation, the route of entry may be considered as nose, nasal cavity. This type of Lakshana can be seen in the patients who has extreme Sahasa (debilitating strenuous exercise). In this case, the patient may have breathing through open mouth. In other words to say, in this situation the entry of the pathogen may occur either through nose or oral cavity which in turn may lead to the manifestation of the symptoms running nose, frequent sneezing. Due to the running nose, there may be a possibility of blockage of nose due to the presence of crust in the nasal cavity. Due to this, the person starts breathing through open mouth which in turn may lead to lack of appetite. All these mechanisms are slow processes in the emaciated individuals.

8 Extreme mucus is produced when there is excess parasympathetic nervous system stimulation. Whenever entry of pathogen occurs in the body, that will lead to the secretions to occur in the system of entry of pathogen. This will lead to excess salivary secretion if the entry of the pathogen is from oral cavity.

9 This type of prodromal symptoms can be seen in case of entry of the pathogens into the alimentary canal. In other words, Rajayakshma due to Sahasa, Vishamashana may develop these kinds of symptoms. For Sahasa, we have discussed above in the context of Purvarupa, refer the context for the meaning. In case of Vishamashana, the person will have abnormal pattern of food intake in terms of quantity of food intake and timing of the food intake. If the person is not having a routine in terms of food intake, following mechanisms may develop in the individuals.

- Lack of secretion of salivary glands leading to dryness of the mouth, as there is lack of functioning of the taste bud, poor taste perception from the individual may occur
- Lack of secretion of gastric juice, inefficient secretion of pepsinogen, inactive form of pepsinogen secretion, poor hydrochloric acid (pH between 1.8 to 3.5 normally, in case of poor gastric juice, pH) may be observed. All these factors may hamper the process of ingestion, storage digestion and assimilation in the alimentary canal.

10 In this, there are two terminologies – swelling in oral cavity and both feet.

- Swelling in the oral cavity: When the person is having poor dental hygiene that may be responsible for the deposition of tartar around the teeth, gums and oral cavity. This may lead to the proliferation of bacteria in and around the teeth, which may in turn damage the teeth, gums and oral cavity. This may lead to the inflammation of oral cavity, teeth, gums, tongue or tonsils. This buccal mucosa (stomatitis) or teeth (dental caries). This may lead to the probable explanation for swelling in the oral cavity.
- Bilateral pedal oedema: When one observes bilateral pedal oedema, indicates circulatory failure where in cardiac dysfunction can be observed. One of the pathological mechanism i.e. hypoproteinemia may be one of the cause of bilateral pedal oedema. This indicates nutritional deficiency in the diseased individual.
- When there is deficiency in the proteins and electrolytes will lead to the decrease in oncotic pressure which leads to the movement of fluid from the capillaries into the interstitial compartment. This could be the main reason for the appearance of bilateral pedal oedema.

11 This premonitory symptom cannot be explained with current understanding.

- ⊗ Whitish discolouration of the eyes¹²
- ⊗ Excess libido¹³
- ⊗ Aversion towards everything¹⁴
- ⊗ Frequent seeing the waterless rivers, empty cities, forest fire etc in dreams
- ⊗ In dreams, he feels as if he is carried by peacock, vulture, monkey, chameleon etc¹⁵.

Sahasajanya Rajayakshma

तत्र साहसं शोषस्यायतनमिति यदुक्तं तदनुव्याख्यास्यामः- यदा पुरुषो दुर्बलो हि सन् बलवता सह विगृह्णाति, अतिमहता वा धनुषा व्यायच्छति, जल्पति वाऽप्यतिमात्रम्, अतिमात्रं वा भारमुद्वहति, अप्सु वा प्लवते चातिदूरम्, उत्सादनपदाघातने वाऽतिप्रगाढमासेवते, अतिप्रकृष्टं वाऽध्वानं द्रुतमभिपतति, अभिहन्यते वा, अन्यद्वा किञ्चिदेवंविधं विषममतिमात्रं वा व्यायामजातमारभते, तस्यातिमात्रेण कर्मणोरः क्षण्यते।

तस्योरः क्षतमुपप्लवते वायुः।

स तत्रावस्थितः श्लेष्माणमुरः स्थमुपसङ्गह्य पित्तं च दूषयन् विहरत्यूर्ध्वमधस्तिर्यक् च।

तस्य योऽंशः शरीरसन्धीनाविशितितेनास्य जृम्भाऽङ्गमर्दो ज्वरश्चोपजायते, यस्त्वामाशयमभ्युपैति तेन रोगा भवन्ति उरस्या अरोचकश्च, यः कण्ठमभिप्रपद्यते कण्ठस्तेनोद्ध्वंस्यते स्वरश्चावसीदति, यः प्राणवहानि स्रोतांस्यन्वेति तेन श्वासः प्रतिश्यायश्च जायते, यः शिरस्यवतिष्ठ ते शिरस्तेनोपहन्यते; ततः क्षणनाच्चैवोरसो विषमगतित्वाच्च वायोः कण्ठस्य चोद्ध्वंसनात् कासः सततमस्य सञ्जायते, स कासप्रसङ्गादुरसि क्षते शोणितं वृद्धिं, शोणितागमनाच्चास्य दौर्बल्यमुपजायते; एवमेते साहसप्रभवाः साहसिकमुपद्रवाः स्पृशन्ति।

ततः स उपशोषणैरेतैरुपद्रवैरुपद्रुतः शनैः शनैरुपशुष्यति।

तस्मात् पुरुषो मतिमान् बलमात्मनः समीक्ष्य तदनुरूपाणि कर्माण्यारभते

कर्तुं; बलसमाधानं हि शरीरं, शरीरमूलश्च पुरुष इति॥४॥

भवति चात्र-

साहसं वर्जयेत् कर्म रक्षद्भीवितमात्मनः।

जीवन् हि पुरुषस्त्विष्टं कर्मणः फलमश्नुते॥५॥

Ca.Ni. - 6/4,5

- ⊗ When a weak person gets confrontation with a physically strong person or when a person excessively indulges in exercise or speaks too much or carries overweight or swims for a long distance in water or forceful anointing and kneading with feet etc.
- ⊗ Excess walking or trauma to the body or abnormal and excess exercise.
- ⊗ The above said measures lead to trauma to the chest and vitiation of Vata in the chest.
- ⊗ This vitiated Vata Dosha will lead to the vitiation of Kapha and Pitta Dosha in the chest and these vitiated Dosha will move in upward, downward and oblique directions.
- ⊗ When the vitiated Dosha enter the joints, then the person will presents with yawning, bodyache and fever.
- ⊗ When the vitiated Dosha enter the Amashaya (stomach), then the individual may suffer from the diseases of Uras (chest) and tastelessness.
- ⊗ When the vitiated Dosha enter the throat, then the person presents with affliction of throat and hoarseness of voice.
- ⊗ When the vitiated Dosha enter the Pranavaha Srotas, then the person presents with breathing discomfort and running nose.
- ⊗ When the vitiated Dosha enter the Shiras (head), then the person presents with diseases of the head.

- 12 Whitish discolouration of the eyes may refer to pallor. The decrease of haemoglobin in the RBC is the one responsible for the pallor. This is indicative of protein deficiency.
- In the patient suffering from Rajayakshma as discussed in the context of Rajayakshma all the essential factors such as macronutrients (carbohydrate, fat and protein) and micronutrients (vitamins such as vitamin A, vitamin B complex and vitamin C, minerals such as sodium, potassium) deficiency.
 - Pallor may be observed as there is decrease in the formation of haemoglobin itself where in alimentary canal is unable to absorb iron, protein, vitamin C, vitamin B12 may be observed.
- 13 Excess libido could be specifically observed in case of Kshayajanya variety of Rajayakshma. If an individual is frequently performing sexual intercourse, that individual will have an urge of performing sexual intercourse frequently.
- 14 This symptom may indicate poor absorption in the alimentary canal. This may lead to the poor peristaltic movement too leading to heaviness of the abdomen.
- To know this word, we have to understand the concept of formation of Ama in the alimentary canal and its movement in the intravascular, interstitial and intracellular compartments.
 - When an individual is having the poor absorption in the stomach due to poor digestive fire (pH of gastric juice >3.5) will be leading to hampered conversion of macromolecules into intermediate and micromolecules.
 - As human beings consume more amount of carbohydrate rich diet, that will lead to the proliferation of pathogenic organisms in the stomach which damage the gastric mucosa.
 - Because of the damage to the gastric mucosa, macromolecules or intermediate molecules will enter into interstitial compartment of the stomach which will enter the intravascular compartment through lymphatic system. Thus, these molecules will be responsible for the poorer movement even in the intravascular, interstitial and intracellular compartments.
 - When there is poor movement in the body due to the macromolecules or intermediate molecules in the interstitial and intracellular compartments, the person may have the presenting complaint of lethargy. This may be the reason for aversion to do every activity by the ailing individual.
- 15 No probable explanation can be possible to this prodromal symptom explained in Rajayakshma.

- When injury occurs to the chest because of abnormal vitiation of *Vata Dosha* and affliction of throat, the person will get continuous coughing, the sequel of this coughing is haemoptysis. The sequel of haemoptysis is debility.
- Thus, the person will get complications gradually due to the progressive worsening of pathology in due course of time.
- That is the reason, one should not do *Sahasa* (excess effort to do any activities more than one's capacities)

Sandharanajanya Rajayakshma

सन्धारणं शोषस्यायतनमिति यदुक्तं तदनुव्याख्यास्यामः- यदा पुरुषो राजसमीपे भर्तुः समीपे वा गुरोर्वा पादमूले द्यूतसभमन्यं वा सतां समाजं स्त्रीमध्यं वा समनुप्रविश्य यानैर्वाऽप्युच्चावचैरभियान् भयात् प्रसङ्गाद्धीमत्त्वाद्धुणित्वाद्वा निरुणद्ध्यागतान् वातमूत्र-पुरीषवेगान् तदा तस्य सन्धारणाद्वायुः प्रकोपमापद्यते, स प्रकुपितः पित्तश्लेष्माणौ समुदीर्योर्ध्वमधस्तिर्यक् च विहरति; ततश्चांशविशेषेण पूर्ववच्छरीरावयवविशेषं प्रविश्य शूलमुपजनयति, भिनन्ति पुरीषमुच्छोषयति वा, पाश्चै चातिरुजति, अंसाववमृद्नाति, कण्ठमुरश्चावधमति, शिरश्चोपहन्ति, कासं श्वासं ज्वरं स्वरभेदं प्रतिश्यायं चोपजनयति; ततः स उपशोषणैरैतैरुपद्रवरूपद्वतः शनैः शनैरुपशुष्यति।

तस्मात् पुरुषो मतिमानात्मनः शरीरेष्वेव योगक्षेमकरेषु प्रयतेत विशेषेण; शरीरं ह्यस्य मूलं, शरीरमूलश्च पुरुषो भवति॥६॥

भवति चात्र- सर्वमन्यत् परित्यज्य शरीरमनुपालयेत्।

तदभावे हि भावानां सर्वाभावः शरीरिणाम्॥७॥

Ca.Ni. - 6/6,7

- When the person suppresses the natural urges of flatus, urine, faecal matter and so on, for example, when an individual is with duties with king or in front of teachers or while playing gambling or in the midst of women (in other words in the middle of opposite sex people) or in the middle of group of people or during travel or due to fear or due to blushfulness or due to disgust
- The above said factors will vitiate *Vata Dosha* and this vitiated *Vata* afflicts the *Pitta* and *Kapha Dosha* and which in turn move in upward, downward and oblique directions.
- When the vitiated *Dosha* enter the different body parts, will lead to the manifestation of pain, formation of unformed stools or constipation, pain in the sides of the chest or pain in the shoulder or sore throat or chest discomfort, headache, cough, breathing discomfort, fever, hoarseness of voice, running nose.
- Complications occur gradually to the body because of gradual depletion of body tissues.
- Therefore, one should maintain proper health of the self.

Kshayajanya Shosha

क्षयः शोषस्यायतनमिति यदुक्तं तदनुव्याख्यास्यामः- यदा पुरुषोऽतिमात्रं शोकचिन्तापरिगतहृदयो भवति, ईर्ष्योत्कण्ठाभयक्रोधादिभिर्वा समाविश्यते, कृशो वा सन् रूक्षान्नपानसेवी भवति, दुर्बल प्रकृतिरनाहारोऽल्पाहारो वा भवति, तदा तस्य हृदयस्थायी रसः क्षयमुपैति; स तस्योपक्षयाच्छोषं प्राप्नोति, अप्रतीकाराच्चाबुध्यते यक्ष्मणा यथोपदेक्ष्यमाणरूपेण॥८॥

यदा वा पुरुषोऽतिहर्षादतिप्रसक्तभावः स्त्रीष्वतिप्रसङ्ग-मारभते, तस्यातिमात्रप्रसङ्गाद्रेतः क्षयमेति।

क्षयमपि चोपगच्छति रेतसि यदि मनः स्त्रीभ्यो नैवास्य निवर्तते, तस्य चातिप्रणीतसङ्कल्पस्य मैथुनमापद्यमानस्य न शुक्रं प्रवर्ततेऽतिमात्रोपक्षीणरेतस्त्वात्, तथाऽस्य वायुर्व्यायच्छमानशरीरस्थैव धमनीरनुप्रविश्य शोणित-वाहिनीस्ताभ्यः शोणितं प्रच्यावयति, तच्छुक्रक्षयादस्य पुनः शुक्रमार्गेण शोणितं प्रवर्तते वातानुसृतलिङ्गम्। अथास्य शुक्रक्षयाच्छोणितप्रवर्तनाच्च सन्ध्यः शिथिलीभवन्ति, रौक्ष्यमुपजायते, भूयः शरीरं दौर्बल्यमाविशति, वायुः प्रकोपमापद्यते; स प्रकुपितो वशिकं शरीरमनुसर्पन्नुदीर्य श्लेष्मपित्ते परिशोषयति मांसशोणिते, प्रच्यावयति श्लेष्मपित्ते संरुजति पाश्चै, अवमृद्नात्यंसी, कण्ठमुद्ध्वंसति, शिरः श्लेष्माणामुपत्वलेश्य प्रतिपूरयति श्लेष्मणा, सन्धीश्च प्रपीडयन् करोत्यङ्गमर्दमरोचकाविपाकी च, पित्तश्लेष्मोत्क्लेशात् प्रतिलोमगत्वाच्च वायुर्ज्वरं कासं श्वासं स्वरभेदं प्रतिश्यायं चोपजनयति; स कासप्रसङ्गादुरसि क्षते शोणितं छीवति, शोणितगमनाच्चास्य दौर्बल्यमुपजायते, ततः स उपशोषणैरैतैरुप-द्रवरूपद्वतः शनैः शनैरुपशुष्यति।

तस्मात् पुरुषो मतिमानात्मनः शरीरमनुरक्षञ्छुक्रमनुरक्षेत्।

परा ह्येषा फलनिर्वृत्तिराहारस्येति॥८॥

भवति चात्र-

आहारस्य परं धाम शुक्रं तद्रक्ष्यमात्मनः।

क्षयो ह्यस्य बहून् रोगान्मरणं वा नियच्छति॥९॥

Ca.Ni. - 6/8,9

- When an individual is afflicted with grief, overthinking, jealousy, anxiety, fear, anger etc
- If an emaciated individual is consuming excess dry food substances and drinks.
- When an individual is having weak *Prakriti*, is on fasting or taking very little amount of food.
- With above said reasons, *Rasa* is going to decrease in *Hridaya*, leading to the manifestation of *Kshaya*, due to neglect at this situation of *Kshaya*, there will be manifestation of *Shosha* and at last, the person will present with *Rajayakshma*.
- When an individual is performing excessive sexual intercourse and due to excess ejaculation, the person will have *Shukra Kshaya* in due course of time. Still the person indulges in sexual intercourse, *Shukra* (semen) will not be ejaculated, in stead, blood will be come during ejaculation. This will be followed by vitiation of *Vata Dosha*.
- Due to the excess loss of semen from the body and excess blood loss from the male urethra due to

excess indulgence in copulation, the person will have looseness of joints, dryness of the body parts, debility of the body.

- Vitiated *Vata Dosha* will circulate all over the body, dries up the *Kapha* and *Pitta* in the *Rakta* and *Mamsa Dhatu*, resulting in the pain in sides of the chest, pain in the shoulder, severe sore throat.
- Vitiated *Kapha* will afflict *Shras* (head), resulting in headache, when it afflicts *Sandhi* (joints), the person develops bodyache, tastelessness and impaired digestion.
- Due to the vitiation of *Pitta* and *Kapha Dosha* and *Pratiloma* (reverse) movement of vitiated *Vata*, the person develops fever, cough, breathlessness, hoarseness of voice and running nose. Due to continuous coughing, the person will have haemoptysis, and due to haemoptysis, the person will have profound debility. Thus, the person will have gradual decline in the health and leading to multiple complications.

Vishamashanajanya Shosha

विषमाशनं शोषस्यायतनमिति यदुक्तं, तदनुव्याख्यास्यामः- यदा पुरुषः पानाशनभक्ष्यलेहोपयोगान् प्रकृतिकरणसंयोगराशि- देशकालोपयोगसंस्थोपशयविषमानासेवते, तदा तस्य तेभ्यो वातपित्तश्लेष्माणो वैषम्यमापद्यन्ते; ते विषमाः शरीरमनुसृत्य यदा स्रोतसामयनमुखानि प्रतिवार्यावतिष्ठन्ते तदा जन्तुर्यद्य-दाहारजातमाहरति तत्तदस्य मूत्रपुरीषमेवोपजायते भूयिष्ठं नान्यस्तथा शरीरधातुः; स पुरीषोपष्टम्भाद्वर्तयति, तस्माच्छुष्यतो विशेषेण पुरीषमनुरक्ष्यं तथाऽन्येषामतिकृशदुर्बलानां; तस्यानाप्यायमानस्य विषमाशनोपचिता दोषाः पृथक् पृथगुपद्रवैर्युञ्जन्तो भूयः शरीरमुपशोषयन्ति।

तत्र वातः शूलमङ्गमर्दं कण्ठोद्ध्वंसनं पार्श्वसंरुजनमंसावमर्दं स्वरभेदं प्रतिश्यायं चोपजनयति; पित्तं ज्वरमतीसारमन्तर्दाहं च; श्लेष्मा तु प्रतिश्यायं शिरसो गुरुत्वमरोचकं कासं च, स कासप्रसङ्गादुरसि क्षते शोणितं निष्ठीवति, शोणितगमनाच्चास्य दौर्बल्यमुपजायते।

एवमेते विषमाशनोपचितास्त्रयो दोषा राजयक्ष्माणमभिनिर्वर्तयन्ति।

स तैरुपशोषणैरुपद्रवैरुपद्रुतः शनैः शनैः शुष्यति।

तस्मात् पुरुषो मतिमान् प्रकृतिकरणसंयोगराशिदेशकालोपयोग- संस्थोपशयादविषममाहारमाहरेत्॥ 10 ॥

भवति चात्र-

हिताशी स्यान्मिताशी स्यात्कालभोजी जितेन्द्रियः।

पश्यन् रोगान् बहून् कष्टान् बुद्धिमान् विषमाशनात्॥ 11 ॥

Ca.Ni. - 6/10,11

- When an individual is taking food substances in edible, drinkable, chewable forms irregularly i.e. not following *Ashta Ahara Vidhi Visheshayatana* (nature, combinations, amount, place, time and rules of dietetics), that will lead to vitiation of *Vata*, *Pitta* and *Kapha*.
- These vitiated *Dosha* will obstruct the different

channels of the body, leading to the formation of *Mutra* and *Purisha* and no other *Dhatu* (body tissues) are formed. That is the reason, when you make the diagnosis of *Rajayakshma*, one should protect the excess movement of *Purisha* from the body.

- Vitiated *Vata Dosha* is responsible for pricking type of pain, bodyache, severe sore throat, pain in the sides of the chest, hoarseness of voice and running nose.
- Vitiated *Pitta Dosha* is responsible for fever, diarrhoea and burning sensation inside the body.
- Vitiated *Kapha Dosha* is responsible for running nose, heaviness of the head, tastelessness and cough. Due to the continuous coughing, the person will have haemoptysis and due to haemoptysis, the patient develops severe debility. Thus, the person will develop the gradual progression of *Rajayakshma*.
- That is the reason, the person has to take adequate amount of food as per the prescribed way.

Lakshana of Rajayakshma based on Ayatana of Rajayakshma

Sahasa	Vega Sandharana	Kshaya	Vishamashana
Shirah Shula	Shirah Shula	Shirah Shula	Shirah Shula
Kanthodh-vamsa	Amsavamarda	Amsa Tapa	Amsa Tapa
Kasa	Kasa	Kasa	Kasa
Svarabheda	Svarabheda	Svarabheda	Svara Bheda
Arochaka	Arochaka	Arochaka	Arochaka
Parshva Shula	Parshva Shula	Parshva Shula	Parshva Shula
Atisara	Atisara	Atisara	Rakta Vamana
Jrimbha	Angamarda	Angamarda	Angamarda
Jvara	Jvara	Jvara	Jvara
Urah Shula	Pratishyaya	Pratishyaya	Pratishyaya
Rakta Mishrita Kapha Shtivana	Vamana	Shvasa	Vamana

Anuloma Kshaya and *Pratiloma Kshaya* are two types of pathological phenomena seen in the patients of *Rajayakshma*. This is the contribution of *Sushruta* and *Madhavakara*.

Anuloma Kshaya is the mechanism where in we will observe sequential loss of body tissues from *Rasa*, *Rakta*, *Mamsa*,

Medas, Asthi, Majja and Shukra.

Pratiloma Kshaya is the mechanism where in we will observe the sequential loss of body tissues in the reverse order i.e. Shukra, Majja, Asthi, Medas, Mamsa, Rakta and Rasa.

Trirupa Rajayakshma¹⁶

अंसपाश्चात्तितापश्च संतापः करपादयोः।

ज्वरः सर्वाङ्गश्चेति लक्षणं राजयक्ष्मणः॥5॥

Ma.Ni. – 10/5

- Amsa Parshvabhitapa – burning sensation of shoulder and sides of the chest
- Karapadayoh Santapa – local raise of temperature on palms and soles
- Sarvangaga Jvara – raised body temperature all over the body

Shadrupi Rajayakshma¹⁷

यक्ष्मणः षडिमानि वा।

कासो ज्वरः पार्श्वशूलं स्वरवर्चोगदोऽरुचिः॥4६॥

- Kasa – cough
- Jvara – fever
- Parshvashula – pricking type of pain in sides of the chest
- Svaragada – hoarseness of voice
- Varcho Gada – diarrhoea
- Aruchi – tastelessness

Ekadasha Rupi Rajayakshma¹⁸

कासोऽसतापो वैस्वर्यं ज्वरः पार्श्वशिरोरुजा।

छर्दनं रक्तकफयोः श्वासवर्चोगदोऽरुचिः॥4५॥

रूपाण्येकादशैतानि यक्ष्मणः

Ca.Ci. – 8/45,46

- Kasa – cough
- Amsatapa – raised temperature in shoulder region
- Vaisvaryā – hoarseness of voice
- Jvara – fever

- Parshvaruja – pain in sides of the chest
- Shiroruja – headache
- Rakta Chardana – vomiting of blood
- Kapha Chardana – vomiting of mucus
- Shvasa – breathlessness
- Varchogada – diarrhoea
- Aruchi – tastelessness

Shosha

Introduction

‘परं दिनसहस्रं तु यदि जीवति मानवः।

सुभिषग्भिरुपक्रान्तस्तरुणः शोषपीडितः’ Madhu, Ma.Ni. – 10/13

संशोषणाद्रसादीनां शोष इत्यभिधीयते।

Su.Ut. – 41/- ; Madhu, Ma.Ni. – 10/1

*Shosha*¹⁹ is the spectrum of the diseases where the suffering individual will be having excess depletion of the body tissues. As per *Madhukosha* commentary, if an individual suffering from *Rajayakshma* and if he lives 1,000 days, then such a patient is considered to be suffering from *Shosha*. According to *Sushruta*, in *Shosha*, we can observe the depletion of *Rasa* etc *Dhatu*, that is why the term *Shosha* is used. As per the *Ayurveda* literatures, *Shosha* is the synonym of *Rajayakshma*. *Madhavakara* and *Sushruta* has mentioned another version of *Shosha* which will be the main discussion topic in the present chapter.

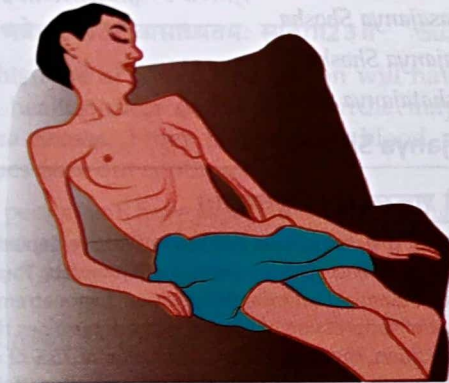


Figure 30: Lakshana of Shosha

- 16 When one observes these Lakshana, it implies that the person may have raise of the temperature especially in the exterior of the body i.e. pathogens may be seen in intravascular compartment.
- 17 Shadrupi Rajayakshma is having 6 major Lakshana – Kasa, Jvara, Parshvashula, Svaragada, Atisara and Aruchi. The presence of Kasa, Jvara, Parshvashula, Svaragada may indicate the affliction of Pranavaha Srotas or respiratory system; where in the presence of Atisara and Aruchi may indicate the affliction of Annavaha Srotas and Purishavaha Srotas or alimentary canal.
 - As per fundamentals of Ayurveda, Mahasrotas is comprised of Pranavaha, Annavaha and Purishavaha Srotas.
 - In this situation, the primary Srotas afflicted in this situation Mahasrotas is afflicted, wherein Pranavaha, Annavaha and Purishavaha Srotas incorporate Mahasrotas.
- 18 When one observe these Lakshana, Kasa, Vaisvaryā, Jvara, Parshvaruja, Shiroruja, Kapha Chardana, Shvasa may indicate the affliction of Pranavaha Srotas; Rakta Chardana, Atisara, Aruchi and Jvara may indicate the Annavaha Srotas affliction. Thus, altogether, Pranavaha, Annavaha and Purishavaha Srotas can be observed. When the person is presented with Jvara, it may indicate the affliction of Rasavaha Srotas.
 - When a clinician connects the dots in relation to Ekadasha Rupi Rajayakshma, we may predict the sequential involvement of Pranavaha, Annavaha, Rasavaha and Purishavaha Srotas.
- 19 Whenever an individual is having depletion of body tissues or emaciation, that implies that the person is having the sequential loss of carbohydrates, fats and proteins from the body. This may also be associated with excess fluid loss and electrolytes as well.

Nidana of Shosha

क्षयाद्वेगप्रतीघातादाघाताद्विषमशानात्॥८॥

जायते कुपितैर्दोषैर्व्याप्त देहस्य देहिनः।

Su.Ut. - 41/8,9

क्षयादित्यादिहेतुभिः कुपितैर्दोषैर्व्याप्तदेहस्य शोषो जायते इति संबन्धः।

Dalhana, Su.Ut. - 41/8,9

- **Dhatu Kshaya**²⁰ (depletion of body tissues)
- Suppression of the **natural urges**²¹
- **Trauma**²²
- Intake of food substances at **abnormal amount and at improper time**²³

As per Dalhana, the person who is exposed to Kshaya etc etiological factors, the vitiated Dosha will move all over the body and lead to the manifestation of Shosha.

Types of Shosha

व्यवायशोकबार्धक्यव्यायामाध्वप्रशोषितान्।

ब्रणोरःक्षतसंज्ञौ च शोषिणौ लक्षणैः शृणु॥१४॥

Ma.Ni. - 10/14

व्यवायशोकस्थाविष्यव्यायामाध्वोपवासतः।

ब्रणोरः क्षत पीडाभ्यां शोषानन्ये वदन्ति हि॥१६॥

Su.Ut. - 41/16

1. Vyavayajanya Shosha
2. Shokajanya Shosha
3. Vardhakya (Sthavirya) janya Shosha
4. Vyayamajanya Shosha
5. Adhvajanya Shosha
6. Upavasajanya Shosha
7. Vranajanya Shosha
8. Urakshatajanya Shosha

Vyavayajanya Shosha

व्यवायशोषी शुक्रस्य क्षयलिङ्गैरुपद्रुतः।

- 20 Because of this, the person will have excessive depletion of the macro nutrients such as carbohydrates, fats and proteins as well as water; and micronutrients such as vitamins and minerals. Thus, we can observe the clinical features related to deficiencies of these substances such as hypoglycaemia; cachexia; dehydration; hyponatremia; vitamin deficiencies etc.
- 21 The type of suppression of natural urges determines the nature of manifestation of Shosha. For example, if the person suppresses the urge of defaecation, that will lead to absorption of 75% of water along with 6.4 to 9.4 litres of air produced in the large intestine, which will be expelled out through the lungs through expiration. If the person is suppressing the urge of coughing, sneezing, effortful breathing; that will lead to reverse entry of pathogens into the lower respiratory tract leading to the manifestation of symptoms of lower respiratory tract.
- 22 When the person is exposed to this particular cause, then first of all, the area of trauma has to be enquired first. If it is externally visible intracellular compartments. Suppose, if the chest injury is being observed, then the person may have severe chest pain, haemoptysis etc lular compartments. Thus area of injury, extent of injury, physical strength of the patient etc are counted while deciding this particular aetiological factor.
- 23 This aetiological factor will lead to the sequential depletion of carbohydrates, fats and proteins, water, minerals, vitamins etc. as the person is not taking the food substances at adequate quantity and at appropriate time too.
- 24 When the individual is indulging in excess sexual intercourse, that may lead to entry of the pathogens through urethra or vagina. As the PHSC (pluripotent haemopoietic stem cells), and these are usually present inside the bone marrow. In due course of time, this may lead to hampered secretion of other cells such as RBCs and platelets.
- 25 When an individual is exposed to grief more frequently, this may lead to activation of sympathetic nervous system which in turn may lead to the hampered secretion of alimentary secretions leading to the decreased absorption of micronutrients to the body.
 - Thus, frequent exposure to grief may lead to sequential loss of carbohydrates, fat and proteins in due course of time.
 - Emaciation will be the sequel in this situation.

पाण्डुर्दो यथापूर्वं क्षीयन्ते चास्य धातवः॥१५॥

Ma.Ni. - 10/15

- If a person indulges in **excess sexual intercourse**²⁴, there will be decrease in **Shukra Dhatu**. Decrease in **Shukra Dhatu** leads to the decrease in **Ojas** and the outcome of this mechanism is manifestation of **Kshaya**. Due to this pathological mechanism, of **Kshaya**. Due to this pathological mechanism, there will be **Pratiloma Kshaya** i.e. first there will be reduction in **Shukra Dhatu**, then the person will have reduced **Majja Dhatu** and this will be responsible for **Asthi Dhatu** and so on till the decrease of **Rasa Kshaya**.
- In this patient, there will be body which will have **Pandu Varna** (yellowish white discolouration).
- In this, the person will have **Shukra Dhatu Kshaya Lakshana**.

“शुक्रक्षये मेढवृषणवेदना, अशक्तिमैथुने, चिराद्वा प्रसेकः, प्रसंक्षाल्यदर्शनं रक्तस्य शुक्रस्य वा”

Madhu, Ma.Ni. - 10/15; Su.Su. - 15/13

- The **Shukra Kshaya Lakshana** mentioned by **Sushruta** are pain in the penis and scrotum, inability to perform sexual activities, delayed ejaculation and presence of blood and little amount of sperm in the semen.

Shoka Shosha

प्रधानशीलः स्वस्ताङ्गः शोकशोष्यपि तादृशः।

Ma.Ni. - 10/16

प्रधानशीलः स्वस्ताङ्गः शोकशोष्यपि तादृशः।

बिना शुक्रक्षयकृतैर्विकारैरभिलक्षितः॥१८॥

Su.Ut. - 41/18

- This type of **Shosha**²⁵ results in an individual due to excess and continuous grief or worry by the patient. The grief may be caused due to sudden departure of the loved ones or a disaster in one's life.
- The person suffering from **Shoka Shosha** is usually suffering from continuous **thinkin** on one particular

point and looseness of the body parts. The person will have depression and he will not have any desire to take food and nutrition.

- Here, the patient will suffer from severe *Dhatu Kshaya* (depletion of body tissues) and he will not suffer from *Shukra Kshaya Lakshana* (clinical features indicating the decrease of *Shukra Dhatu* in the body).

Jara (Sthaviirya or Vardhakya) Shosha²⁶

जराशोषी कृशो मन्दवीर्यबुद्धिबलेन्द्रियः॥१६॥

कम्पनोऽरुचिमान् भिन्नकांस्यपात्रहतस्वरः।

तिष्णणा हीनं गौरवारतिपीडितः॥१७॥

संप्रसृतास्यनासाक्षिः शुष्करूक्षमलच्छविः।

Ma.Ni. – 10/16,17; Su.Ut. – 41/19,20

- This type of *Shosha* is seen in elderly individuals. The person will be emaciated and there will be vitiation of *Vata Dosha*. He will have less physical strength, less mental stamina and poor functioning of *Indriya*. Apart from the above said *Lakshana*, following *Lakshana* may also be observed.
- Involuntary movements.
- Tastelessness
- Loss of voice or the sound produced similar to the sound produced by a broken bronze vessel
- Cough with less sputum
- Heaviness of the body parts and restlessness
- Deformities of mouth, nose and eyes
- Dry faecal matter and sometimes very hard stools

Adhva Shosha²⁷

अध्वशोषी च स्वस्ताङ्गः संभृष्टपरुषच्छविः॥

प्रसृतागात्रावयवः शुष्कक्लोमगलाननः।

Ma.Ni. – 10/18,19; Su.Ut. – 41/21

- In this type of *Shosha*, the main cause is excess walking. Excess walking causes *Mamsa Kshaya* which in turn can be identified by depletion of muscular tissue.

26 In this situation, the person may have one or multiple debilitating illnesses, which may lead to the sequential depletion of the carbohydrates, fats and proteins. This may lead to the metabolic acidosis in the body (ketoacidosis and lactic acidosis). Thus, the person may have emaciation of the body.

27 The term *Adhva* refers to walking. In this context, the term excess walking more than one's capacity has to be considered. In a marathon race (running for a distance of 42.2 kms/ 26.2 miles), the metabolism of the body may increase to 2000% above normal metabolism at rest. In other words, there will be sequential depletion of carbohydrates, fats and proteins, and electrolytes can be seen.

28 In this instance, the term *Vyayama* refers to excess exercise more than one's own capacity. This can be understood by knowing the rate of metabolism occurring in marathon race. In a marathon race (running for a distance of 42.2 kms/ 26.2 miles), the metabolism of the body may increase to 2000% above normal metabolism at rest. In other words, there will be sequential depletion of carbohydrates, fats and proteins, and electrolytes can be seen.

29 When an individual presents with non healing ulcer for a longer period of time, that will lead to the rapid depletion of macronutrients (carbohydrates, fats and proteins) and micronutrients (electrolytes). Thus, in this situation, the suffering individual may have emaciation of the body.

30 In fasting, the person may have sequential depletion of macronutrients carbohydrates, fats and proteins respectively, along with water and electrolytes sodium, potassium, chloride and calcium ions in particular. Thus, the person who are on continuous fasting for longer time period, there may be the possibility of emaciation of the body.

- The person will present with poor functioning of body parts
- Dry and lustreless skin
- Numbness of tingling sensation of the skin
- Dryness of the palate, throat and oral cavity

Vyayama Shosha²⁸

व्यायामशोषी भूयिष्ठमेभिरेव समन्वितः।

लिङ्गैरुःक्षतकृतैः संयुक्तश्च क्षतं विना॥१९॥

Ma.Ni. – 10/19

व्यायामशोषी भूयिष्ठमेभिरेव समन्वितः।

उरःक्षतकृतैर्लिङ्गैः संयुक्तश्च क्षताद्विना॥२०॥

Su.Ut. – 41/22

- In this type of *Shosha*, the *Lakshana* mentioned in *Adhva Shosha* will be manifested in severe form. This manifests in an individual who does excessive physical activity beyond one's capacity.
- There will be decrease of *Medas* and *Mamsa* which leads to the loss of weight.
- The *Lakshana* of *Vyayama Shosha* are similar to *Adhva Shosha*.

Vrana Shosha²⁹

रक्तक्षयाद्वेदनाभिस्तथैवाहारयन्त्रणात्।

व्रणितस्य भवेच्छोषः स चासाध्यतमो मतः॥२०॥

Ma.Ni. – 10/20

रक्तक्षयाद्वेदनाभिस्तथैवाहारयन्त्रणात्।

व्रणितस्य भवेच्छोषः स चासाध्यतमः स्मृतः॥२१॥

Su.Ut. – 41/23

- In this type of *Shosha*, the person will have chronic non healing ulcer or fistula, then he may develop *Vrana Shosha*. From the wound, blood and other tissues ooze out continuously.
- The person will have associated with *Lakshana* such as severe pain at the wound region and it is considered to be *Asadhya* (incurable).

Upavasajanya Shosha³⁰

Only the terminology of *Upavasajanya Shosha* is mentioned

and the description of *Upavasajanya Shosha* is not being described in either *Sushruta Samhita* or *Madhava Nidana*.

Urakshatajanya Shosha³¹

व्यायामभाराध्ययनैरभिघातातिमैथुनैः।

कर्मणा चाप्युरस्येन वक्षो यस्य विदारितम्।

तस्योरसि क्षते रक्तं पूयः श्लेष्मा च गच्छति॥24॥

कासमान छर्दयेच्च पीतरक्तासितारुणम्।

संतप्तवक्षाः सोऽत्यर्थं दूयनात्परिताम्यति॥25॥

दुर्गन्धवदनोच्चासो भिन्नवर्णखरो नरः॥26॥

Su.Ut. – 41/24-26

- The etiological factors such as excess exercise, lifting heavy weights, excess studying, trauma, excess

sexual intercourse will lead to the injury to the chest. This is called as *Urakshatajanya Shosha*.

- This is associated with trauma to chest and the person passes reddish, pus mixed, mucoid sputum on coughing and sometimes, he will have vomiting with yellowish, reddish or whitish coloured content.
- He may have burning sensation in the chest, foul smell from the mouth and foul smell from the breath and hoarseness of voice.
- The description of *Urakshatajanya Shosha* given by *Charaka* is similar to the description of *Urakshata* mentioned in the context of *Kshatakshina Adhyaya*.

Key summary of the chapter

- Rajayakshma is the spectrum of diseases where in vitiation of all three Dosha vitiation, affliction of all seven Dhātu is being observed.

अनेकरोगानुगतो बहुरोगपुरोगमः। राजयक्ष्मा क्षयः शोषो रोगराडिति च स्मृतः॥1॥

A.Hr.Ni. – 5/1

- Rajayakshma is considered to be associated with multiple diseases which occur prior to the manifestation of Rajayakshma and multiple diseases will follow Rajayakshma. That is the reason, Rajayakshma is considered to be the king of the diseases or one of the major diseases.
- As far as types of Rajayakshma are concerned, 4 types of Rajayakshma are mentioned based on Ayatana or Nidana. They are *Sahasajanya*, *Vishamashanajanya*, *Sandharanajanya* and *Kshayajanya*.
- Depending upon the pathogenesis happening inside the patient of Rajayakshma, two types of *Samprapti* are being mentioned. *Anuloma Kshaya* and *Pratiloma Kshaya*. In case of *Anuloma Kshaya*, sequential depletion of Dhātu from *Rasa* to *Shukra Dhātu* occurs. In case of *Pratiloma Kshaya*, sequential depletion of Dhātu from *Shukra* to *Rasa Dhātu* occurs.
- Depending upon the nature of symptoms of Rajayakshma, three types of Rajayakshma are mentioned. They are *Trirupi* (having 3 main Lakshana), *Shadrupi* (having 6 main Lakshana) and *Ekadasharupi* (having 11 main Lakshana) Rajayakshma.
- *Pranavaha*, *Annavaha*, *Purishavaha* and *Rasavaha Srotas* affliction may be seen initially, later affliction of multiple Srotas can be observed.
- Rajayakshma is considered to be associated with multiple diseases which occur prior to the manifestation of Rajayakshma and multiple diseases will follow Rajayakshma. That is the reason, Rajayakshma is considered to be the king of the diseases or one of the major diseases.

‘परं दिनसहस्रं तु यदि जीवति मानवः। सुभिषग्भिरुपक्रान्तस्तरुणः शोषपीडितः’
संशोषणाद्रसादीनां शोष इत्यभिधीयते।

Madhu, Ma.Ni. – 10/13

Su.Ut. – 41/- ; Madhu, Ma.Ni. – 10/1

- There are two different meanings for the description *Shosha*. As per *Madhukosha* commentary, if an individual is suffering from Rajayakshma and if he lives his/ her life 1,000 days, then such individual is said to be suffering from *Shosha* disease. As per *Sushruta*, depletion of *Rasa* etc Dhātu is observed in *Shosha*. In essence, *Shosha* is that is the synonym of Rajayakshma.
- *Dhatukshaya* (depletion of body tissues); suppression of natural urges; trauma and intake of food at inappropriate quantity and at improper time are the Nidana for the manifestation of *Shosha*.
- 8 types of *Shosha* are explained in *Madhava Nidana* by *Madhavakara*. They are *Vyavayajanya*; *Shokajanya*; *Vardhakyajanya*; *Vyayamajanya*; *Adhvajanya*; *Upavasajanya*; *Vranajanya*; *Urakshatajanya Shosha*.

G K 31 When an individual is presented with chest injury, there may be the possibility of the affliction of rib fractures, aortic ruptures, lung contusion or development of shock due to internal haemorrhages.

- This condition may be associated with rapid depletion of macronutrients and micronutrients as this is one of the grave condition affecting the individuals.
- Thus, the person may have emaciation secondary to the development of metabolic acidosis (ketoacidosis and lactic acidosis).

Probable questions asked in university examination

Short essay questions

- Explain the Sankhya Samprapti of Rajayakshma
- Explain Sahasajanya Rajayakshma
- Explain Ekadasharupi Rajayakshma
- Explain the types of Samprapti of Rajayakshma
- Explain the types of Shosha
- Describe Nidana and types of Shosha

MCQs

1. Which of the following is not the synonym of Rajayakshma?
- Shosha;
 - Kshaya;
 - Rogarat;
 - Varchogada

Answer: (d)

2. The Dhatu Kshaya begins in Pratiloma Kshaya type of Rajayakshma from
- Rasa Dhatu;
 - Majja Dhatu;
 - Asthi Dhatu;
 - Shukra Dhatu

Answer: (d)

3. Which of the following is not the Lakshana of Shadrupi Rajayakshma?
- Rakta Chardana;
 - Shvasa;
 - Atisara;
 - Kasa

Answer: (a)

4. Sankhya Samprapti of Rajayakshma is described based on
- Lakshana;
 - Nidana;
 - Purvarupa;
 - Upashaya

Answer: (b)

5. Which of the following is not mentioned in the context of Sankhya Samprapti of Rajayakshma?
- Sahasa;
 - Vishamashana;
 - Vegasandharana;
 - Adhyashana

Answer: (d)

6. Find the one which does not belong to one group
- Carbohydrate;
 - Glucose;
 - Fat;
 - Protein

Answer: (b)

7. Mechanism of ketoacidosis occurs as a result of over-metabolism of
- Carbohydrate;
 - Fat;
 - Protein;
 - Myoglobin

Answer: (b)

8. The terms Anuloma and Pratiloma are associated with which of the following diseases?

- Shvasa;
- Hikka;
- Kasa;
- Rajayakshma

Answer: (d)

9. Langhana therapy is indicated in all of the diseases except

- Jvara;
- Raktapitta;
- Kaphaja Gulma;
- Rajayakshma

Answer: (d)

10. Rogamarga afflicted in the Rajayakshma disease is

- Bahya;
- Abhyantara;
- Madhyama;
- None of the above

Answer: (c)

11. The number of days a person will live in case of diagnosis of Shosha from the day of manifestation as per Madhukosha commentary is

- 100 days;
- 500 days;
- 1,000 days;
- 10,000 days

12. Sankhya Samprapti of Shosha is

- 5 types;
- 7 types;
- 8 types;
- 1 type

Answer: (c)

13. The word Sthavira in Sthavirajanya Shosha refers to Shosha observed in

- Children;
- Middle aged males;
- Females of reproductive age group;
- Old aged people

Answer: (d)

14. Scurvy is caused by the deficiency of

- Vitamin A;
- Vitamin C;
- Vitamin D;
- Vitamin E

Answer: (b)

15. The description of Urakshatajanya Shosha resembles the description of of Charaka Samhita.

- Kasa;
- Shvasa;
- Kshatakshina;
- Svarabheda

Answer: (c)

Chapter resources

- <https://my.clevelandclinic.org/health/diseases/24492-metabolic-acidosis>
- <https://wellcomecollection.org/works/hur7rekt/items>
- Charaka Samhita, Nidana Sthana – 6th chapter – Shosha Nidana Adhyaya

- Charaka Samhita, Chikitsa Sthana - 8th chapter - Rajayakshma Chikitsa Adhyaya
- Sushruta Samhita, Uttara Tantra - 41st chapter - Shosha Pratishedha Adhyaya
- Ashtanga Hridaya, Nidana Shtana - 5th chapter - Rajayakshma Nidana Adhyaya
- Madhava Nidana - 10th chapter - Rajayakshma Kshatakshina Nidana Adhyaya
- Charaka Samhita, Nidana Sthana 6th chapter - Shosha Nidana Adhyaya
- Sushruta Samhita, Uttara Tantra - 41st chapter - Shosha Pratishedha Adhyaya
- Madhava Nidana 10th chapter - Rajayakshma Kshatakshina Nidana
- Golwalla's medicine for students by Aspi F Golwalla

- and Sharukh A Golwalla, edited by Milind Y Nadkar, 25th edition, Jaypee Brothers Medical Publishers (P) Ltd, New Delhi, 2017
- The Merck Manual of Diagnosis and Therapy by Robert S Porter and Justin L Kaplan, 19th edition, Merck Sharp & Dohme Corp, New Jersey, 2011
- Chamberlain's symptoms and signs in clinical medicine by Colin Ogilvie and Christopher C Evans, 12th edition, Butterworth-Heinemann Publishers, 1997
- Best & Taylor's physiological basis of medical practice by O P Tandon and Y Tripathi, 13th edition, Wolters Kluwer Health (India) Pvt. Ltd, Gurgaon, 2012
- Textbook of Medical physiology by Arthur C Guyton and John E Hall, 11th edition, Saunders Elsevier publications, Philadelphia, 2006
